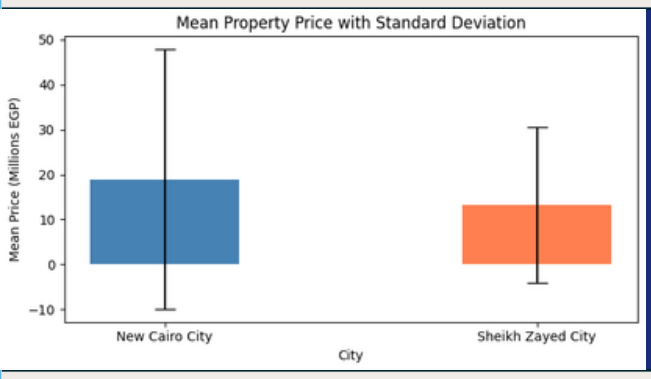
A Comparative Statistical Analysis of Residential Property Pricing

Project 3



PROJECT OVERVIEW

This project analyzes approximately 20,000 real estate listings from Property Finder Egypt to determine if a statistically significant difference in residential property prices exists between New Cairo City and Sheikh Zayed City.

DATA INGESTION & PREPROCESSING

Targeted Extraction: Imported the raw real estate dataset, utilizing Pandas and NumPy, filtering the dataframe to strictly isolate records for 'New Cairo City' and 'Sheikh Zayed City'.

Feature Cleaning: Applied string replacement and regular expressions expressions to strip non-numeric characters (like commas) from the price column and extract pure numeric float values from the size strings.

Missing Values: Records with missing price entries were removed using listwise deletion, ensuring only verified price points were used for hypothesis testing.

STATISTICAL METRICS & SEGMENTATION

MarketSegmentation: Bifurcated the fullycleaned datasetinto two distinct, standalone dataframes representing our primary independent groups: New Cairo and Sheikh Zayed.

Central Tendency: Computed the exact statistical Mean and Median for property prices within both geographic subsets to establish the baseline market valuations.

Variance Analysis: Calculated the Standard Deviation for both regions to quantify market volatility and visually understand the spread of property pricing around the average in each city.

RESEARCH HYPOTHESES

Testing at a 0.05 significance level utilizing an independent two-sample t-test. Null Hypothesis (H0): The average property price in New Cairo is equal to Sheikh Zayed. Alternative Hypothesis (H1): The average property prices significantly differ between the two cities.



Yalla-Nanalyze

GitHub

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OUTCOMES & INFRASTRUCTURE

Projections rejecting the null hypothesis, revealing a distinct price premium in New Cairo driven by its proximity to the New Administrative Capital.